

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Scenario-Based Assignment (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Assignment
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:		Date:	06/08/2026

Code of Conduct

I Thanmai Reddy (192425372) (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____

QUESTIONS: 5

Total Marks: 50

Annexure A

Scenario-Based Assignment

Duration: 1hr

Deep Learning – Assessment 2

1. Machine Learning vs Deep Learning

A healthcare company wants to develop an AI system to detect pneumonia from chest X-ray images. Initially, the development team plans to use traditional Machine Learning with manually extracted image features. However, after collecting millions of medical images, they decide to switch to a Deep Learning approach.

Question:

Compare Machine Learning and Deep Learning in the context of this scenario. Justify why Deep Learning is a better choice for this application by discussing data requirements, feature extraction, computational resources, accuracy, and real-world implementation challenges.

A healthcare company wants to detect pneumonia from chest X-ray images. Initially, it uses **Machine Learning (ML)** with manually extracted image features. After collecting millions of X-ray images, the company switches to **Deep Learning (DL)** because it is more suitable for large-scale image analysis.

Machine Learning (ML):

- Works well with small or medium-sized datasets.
- Requires manual feature extraction, where experts identify features such as edges, texture, and shape.
- Can be trained using normal CPUs with lower computational cost.
- Accuracy depends on the quality of manually selected features.
- Manual feature engineering is time-consuming and may miss important disease patterns.

Deep Learning (DL):

- Requires a very large dataset, such as millions of X-ray images.
- Automatically learns meaningful features directly from raw images without manual feature extraction.
- Requires powerful GPUs or TPUs and more computational resources.
- Provides higher accuracy by learning complex image patterns.
- Needs more training time and labeled data but performs much better for medical image analysis.

Justification

Deep Learning is a better choice because the healthcare company has collected millions of chest X-ray images. It can automatically learn complex features from the images, reducing the need for manual feature engineering. Deep Learning models, especially **Convolutional Neural Networks (CNNs)**, achieve higher accuracy and are widely used in real-world medical diagnosis. Although they require more computational power, larger datasets, and longer training time, their superior performance makes them the preferred choice for pneumonia detection.

2. Representation Learning

A retail company has millions of customer purchase records and wants to automatically identify customer buying patterns without manually selecting important features. The company plans to use a Deep Learning model that can learn meaningful representations directly from raw transactional data.

Question:

Explain the concept of Representation Learning and discuss how it helps solve the company's problem. Compare it with traditional feature engineering and explain its advantages in handling complex real-world datasets.

Representation Learning is a technique in which a model automatically learns useful and meaningful features directly from raw data instead of relying on manually designed features. In Deep Learning, the model discovers hidden patterns and relationships in the data that are useful for making predictions.

In this scenario, the retail company has millions of customer purchase records. Instead of manually selecting features such as purchase frequency, average spending, or product categories, the Deep Learning model automatically learns customer buying patterns from the raw transactional data. This helps the company better understand customer behavior and improve tasks such as product recommendations, customer segmentation, and sales prediction.

In traditional feature engineering, domain experts manually identify and create important features before training the model. This process is time-consuming, requires expert knowledge, and may overlook hidden relationships in the data. The quality of the final model depends heavily on the manually selected features.

Representation Learning overcomes these limitations by automatically extracting the most relevant features during training. It can capture complex and non-linear relationships that are difficult to identify manually. It also scales well to very large datasets and reduces the need for human effort.

Advantages of Representation Learning:

- Automatically learns useful features from raw data.
- Eliminates the need for manual feature engineering.
- Captures complex and hidden patterns in large datasets.
- Improves prediction accuracy and model performance.
- Handles high-dimensional and real-world data efficiently.
- Saves time and reduces dependence on domain experts.

Therefore, Representation Learning is an ideal approach for the retail company because it can efficiently analyze millions of customer transactions, discover meaningful buying patterns, and provide more accurate business insights than traditional feature engineering.

3. Width and Depth of Neural Networks

A startup is developing a handwritten digit recognition system. Two engineers propose different neural network architectures. One suggests using a very wide network with a single hidden layer containing thousands of neurons, while the other recommends a deeper network with multiple hidden layers and fewer neurons per layer.

Question:

Analyze both approaches by explaining the concepts of width and depth in neural networks. Discuss how each architecture affects feature learning, computational complexity, model performance, and generalization. Recommend the most suitable architecture with proper justification.

A neural network's **width** refers to the number of neurons in each hidden layer, while its **depth** refers to the number of hidden layers in the network. Both influence how the network learns from data.

A **wide neural network** has a single hidden layer with a large number of neurons. It can learn many patterns at the same level and is suitable for solving simple problems. However, it requires a large number of parameters, increases memory usage, and may overfit the training data, reducing its ability to perform well on new data.

A **deep neural network** contains multiple hidden layers with fewer neurons in each layer. Each layer learns different levels of features. The initial layers learn simple features such as edges and lines, while deeper layers learn more complex features such as shapes and complete digits. This hierarchical learning makes deep networks highly effective for image recognition tasks.

A wide network generally has lower training complexity because it has fewer layers, but it may require more neurons and memory. A deep network takes more time to train due to multiple layers but uses its parameters more efficiently and can learn complex patterns with better accuracy.

In terms of **model performance**, wide networks perform well on simple tasks but may struggle to capture complex relationships. Deep networks usually achieve higher accuracy because they learn features at multiple levels. They also provide better **generalization**, meaning they perform well on unseen data and reduce the risk of overfitting when trained with sufficient data and proper techniques.

Recommendation:

A **deep neural network with multiple hidden layers and fewer neurons per layer** is the most suitable architecture for handwritten digit recognition. It automatically learns simple and complex image features in different layers, provides higher accuracy, improves generalization, and is widely used in modern image recognition systems such as handwritten digit classification. Therefore, the second engineer's approach is the better choice.

4. Activation Functions

A company is building a Deep Learning model for image classification. During training, the team observes problems such as slow convergence, vanishing gradients, dead neurons, and unstable outputs in the final classification layer.

Question:

Evaluate the suitability of Sigmoid, Tanh, ReLU, Leaky ReLU, ELU, and Softmax activation functions for different layers of the neural network. Explain which activation function should be used in each stage of the model and justify your choices based on their characteristics.

Activation functions help a neural network learn complex patterns by introducing non-linearity. Different activation functions are suitable for different layers depending on the problem.

Sigmoid:

The Sigmoid activation function produces output values between **0 and 1**, making it useful for binary classification output layers. However, it suffers from the **vanishing gradient problem**, which slows down learning in deep networks. Therefore, it is generally not recommended for hidden layers.

Tanh:

The Tanh activation function produces output values between **-1 and 1**. Since its output is zero-centered, it performs better than Sigmoid during training. However, it also suffers from the vanishing gradient problem when used in deep networks.

ReLU(RectifiedLinearUnit):

ReLU outputs zero for negative values and the input itself for positive values. It trains faster, reduces the vanishing gradient problem, and is the most commonly used activation function for hidden layers. However, some neurons may stop learning permanently, causing the **dead neuron problem**.

LeakyReLU:

Leaky ReLU is an improved version of ReLU. It allows a small non-zero output for negative values, preventing dead neurons. It is a better choice than ReLU when dead neuron problems occur.

ELU(ExponentialLinearUnit):

ELU also allows negative outputs but produces smoother values than Leaky ReLU. It improves learning speed, reduces vanishing gradients, and provides more stable training. It is suitable for deep neural networks where stable convergence is important.

Softmax:

Softmax converts the outputs of the final layer into probabilities whose sum equals 1. It is used only in the **output layer** for multi-class classification problems, such as image classification, because it predicts the probability of each class.

Recommended Activation Functions

- **Hidden Layers:** Use **ReLU** because it trains quickly and reduces the vanishing gradient problem.
- **If dead neurons occur:** Replace ReLU with **Leaky ReLU**.
- **For more stable and faster training:** Use **ELU** as an alternative to ReLU.
- **Output Layer:** Use **Softmax** because the task is multi-class image classification and Softmax produces class probabilities.
- **Sigmoid** is suitable only for binary classification output layers.
- **Tanh** can be used in some hidden layers but is less preferred than ReLU-based functions in deep networks.

Justification

For this image classification model, **ReLU (or Leaky ReLU/ELU)** should be used in the hidden layers because they reduce vanishing gradients and improve training speed. **Softmax** should be used in the final output layer because it provides the probability of each image class, making it the best choice for multi-class classification.

5. Unsupervised Training of Neural Networks

A social media platform collects billions of unlabeled user images every month. Since manually labeling such a massive dataset is expensive and time-consuming, the company wants to train a neural network that can automatically discover hidden patterns and useful representations.

Question:

Explain how unsupervised training of neural networks can address this problem. Discuss the learning process, advantages over supervised learning in this scenario, and describe suitable techniques that can be applied for representation learning.

Unsupervised training is a learning approach in which a neural network learns from **unlabeled data** without requiring predefined output labels. The model automatically discovers hidden patterns, structures, and meaningful representations from the input data.

In this scenario, the social media platform collects **billions of unlabeled user images** every month. Since manually labeling such a huge dataset is expensive and time-consuming, unsupervised learning allows the neural network to learn useful image features automatically. These learned representations can later be used for tasks such as image classification, image search, recommendation systems, and face recognition.

The learning process involves finding similarities and differences among images without knowing their categories. The model groups similar images together and learns important visual features such as edges, shapes, textures, and objects. Popular techniques used for unsupervised training include **Autoencoders**, **Restricted Boltzmann Machines (RBMs)**, **Self-Organizing Maps (SOMs)**, and **Generative Adversarial Networks (GANs)**.

Advantages of Unsupervised Training:

- Does not require labeled data, reducing labeling cost and effort.
- Automatically discovers hidden patterns and meaningful features.
- Can efficiently process very large datasets.
- Learns useful representations that improve the performance of supervised tasks.
- Helps in clustering, anomaly detection, image retrieval, and recommendation systems.
- Scales well to real-world applications with massive amounts of data.

Conclusion:

Unsupervised training is the best solution for the social media platform because it can learn meaningful representations from billions of unlabeled images without manual annotation. This reduces cost, improves scalability, and enables the development of accurate AI systems for various image-related applications

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____